

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250286942

Kind Code

A1

Publication Date

September 11, 2025

Inventor(s)

QIU; Zutao et al.

CONNECTION STRUCTURE AND PORTABLE STORAGE DEVICE

Abstract

A connection structure includes: a housing; an interface member, where at least a portion of the interface member extends from a first side wall of the housing to an outside of the housing to be connected with an interface of an electronic device; and a blocking portion, where the blocking portion is arranged on the first side wall of the housing, the blocking portion extends from the first side wall towards the outside of the housing and surrounds an intersection of the interface member and the first side wall.

Inventors: QIU; Zutao (SHENZHEN, CN), LIN; Wei (SHENZHEN, CN), LIU; Quangui (SHENZHEN, CN), ZHOU; Xiping (SHENZHEN, CN)

Applicant: Lexar Electronics (Shenzhen) Co., Ltd. (SHENZHEN, CN)

Family ID: 93793048

Assignee: Lexar Electronics (Shenzhen) Co., Ltd. (SHENZHEN, CN)

Appl. No.: 19/071049

Filed: March 05, 2025

Foreign Application Priority Data

CN 202420436335.7

Mar. 06, 2024

Publication Classification

Int. Cl.: H04M1/02 (20060101); H01R31/06 (20060101)

U.S. Cl.:

CPC H04M1/0274 (20130101); H01R31/06 (20130101); H01R2201/16 (20130101)

Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS

[0001] The present application claims the priority of the Chinese patent application No. 202420436335.7, filed on Mar. 6, 2024, the entire contents of which are hereby incorporated by reference in their entireties.

TECHNICAL FIELD

[0002] The present disclosure relates to the technical field of electrics, and in particular to a connection structure and a portable storage device.

BACKGROUND

[0003] A portable storage device, such as a portable solid-state drive, has a small size, can be portable easily, and has a large storage capacity.

[0004] In order to protect a mobile phone, a user usually cover the mobile phone with a phone case. In the art, an interface member of the portable storage device may be excessively long. In this case, as shown in FIG. 1, when the portable storage device is used with the mobile phone covered with the phone case, and when a connecting structure in the art having the excessively long interface member is connected to an interface of the mobile phone covered with the phone case, a portion of the interface member may be exposed to an outside, protection for electrostatic discharge (ESD) and electromagnetic compatibility (EMC) may be weakened, and an outer appearance is displeased. Alternatively, in the art, a male connector of the portable storage device may be configured to be excessively short, as shown in FIG. 2, which is a schematic view of the connection structure in the art having the excessively short interface member being connected to the interface of the mobile phone covered with the phone case. As shown in FIG. 2, when the male connector is excessively short, poor signal connection may be caused.

SUMMARY OF THE DISCLOSURE

[0005] The present disclosure provides a connection structure and a portable storage device, to solve the above technical problems.

[0006] In a first aspect, the present disclosure provides a connection structure, including: a housing; an interface member, wherein at least a portion of the interface member extends from a first side wall of the housing to an outside of the housing to be connected with an interface of an electronic device; and a blocking portion, wherein the blocking portion is arranged on the first side wall of the housing, the blocking portion extends from the first side wall towards the outside of the housing and surrounds an intersection of the interface member and the first side wall.

[0007] In some embodiments, a height of the blocking portion extending towards the outside of the housing is not less than 1.6 millimeters.

[0008] In some embodiments, the height of the blocking portion extending towards the outside of the housing is 1.6 millimeters.

[0009] In some embodiments, the interface member is one of: a Type-C interface, a lightning interface, a Micro USB interface, and a MiniUSB interface.

[0010] In some embodiments, the interface member is the Type-C interface, the interface member comprises a snapping segment that is not surrounded by the blocking portion, the snapping segment has an extension height of not less than 6.5 millimeters.

[0011] In some embodiments, a width of the blocking portion is less than 12 millimeters.

[0012] In some embodiments, a size of the blocking portion is progressively reduced along a direction of extending towards the outside of the housing.

[0013] In some embodiments, connection between a second side wall of the blocking portion and a surface of the blocking portion away from the housing is a curved surface.

[0014] In some embodiments, the blocking portion and the housing are configured as an integral

and one-piece structure; the blocking portion is formed by a portion of the first side wall of the housing protruding outwardly.

[0015] In a second aspect, the present disclosure provides a portable storage device, including: a body portion; and the connection structure as described in the above. The body portion is arranged inside the housing of the connection structure.

[0016] According to the present disclosure, the connection structure and the portable storage device are provided. The connection structure includes the housing, the interface member, and the blocking portion. At least a portion of the interface member extends from a first side wall of the housing to the outside of the housing to be connected with the interface of the electronic device. The blocking portion is arranged on the first side wall of the housing. The blocking portion extends from the first side wall to the outside of the housing and surrounds the intersection of the interface member and the first side wall. The blocking portion is disposed surrounding the intersection of the interface member and the first side wall of the connection structure. When the connection structure is connected to the mobile phone covered with the phone case, the portion of the interface member not surrounded by the blocking portion is inserted in the interface of the mobile phone. The region of the interface member at the opening of the phone case is blocked by the blocking portion, preventing the interface member from being exposed at the opening. In this way, exposure of the interface member, due to the interface member being excessively long, may be avoided; and poor signal connection, due to the interface member being excessively short, may be avoided.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0017] In order to more clearly illustrate technical solutions in embodiments of the present disclosure, accompanying drawings used for describing the embodiments will be briefly introduced in the following. Apparently, the accompanying drawings in the following description show only some of the embodiments of the present disclosure, and other accompanying drawings may be obtained according to the following drawings by any ordinary skilled person in the art without making any creative work.

[0018] FIG. 1 is a structural schematic view of the connection member in the art, when having the excessively long interface member, being connected to the interface of the mobile phone covered with the phone case.

[0019] FIG. 2 is a structural schematic view of the connection member in the art, when having the excessively short interface member, being connected to the interface of the mobile phone covered with the phone case.

[0020] FIG. 3 is a structural schematic view of a connection structure according to an embodiment of the present disclosure.

[0021] FIG. 4 is a structural schematic view of the connection structure being connected with an interface of a mobile phone that is uncovered with a phone case, according to an embodiment of the present disclosure.

[0022] FIG. 5 is a structural schematic view of the connection structure being connected with the interface of the mobile phone that is covered with the phone case, according to an embodiment of the present disclosure.

[0023] Reference numerals in the drawings: **10**, connection structure; **11**, housing; **111**, first side wall; **12**, interface member; **121**, snapping segment; **13**, blocking portion; **131**, surface; **132**, second side wall; **20**, mobile phone; **30**, phone case.

DETAILED DESCRIPTION

[0024] The technical solutions in the embodiments of the present disclosure will be described clearly and completely in the following by referring to the accompanying drawings in the

embodiments of the present disclosure. Apparently, the described embodiments are only a part of, not all of, the embodiments of the present disclosure. All other embodiments, which are obtained by any ordinary skilled person in the art based on the embodiments in the present disclosure without making creative work, shall be within the scope of the present disclosure.

[0025] Terms used in the embodiments of the present disclosure are used merely for the purpose of describing particular embodiments and are not intended to limit the present disclosure. Any singular form of “a”, “said”, and “the” used in the embodiments and in the claims are intended to include a plural form of a structure, unless clearly indicated in the context. “A plurality of” generally means at least two, but does not preclude a situation that the number of structures is at least one.

[0026] It should be understood that the term “and/or” used herein is merely a description of an association relationship of objects, indicating that three types of relationships may exist. For example, A and/or B may mean that A is present alone, both A and B are present, and B is present alone. In addition, the character “/” herein indicates that an object before the character “or” an object after the character.

[0027] It should be understood that terms “include”, “comprise”, or any other variant used herein are intended to encompass non-exclusive inclusion, such that a process, a method, an article or an apparatus includes not only the listed elements but also other elements that are not explicitly listed or elements that are inherently included in the process, the method, the article or the apparatus. Without further limitation, an element defined by a phrase “including . . .” does not exclude existence of another identical element included in the process, the method, the article, or the apparatus.

[0028] When a portable storage device is used with a mobile phone covered with a phone case, usage of the portable storage device may be affected when an interface member of the portable storage device is excessively long or excessively short.

[0029] Accordingly, the present disclosure provides a connection structure and a portable storage device, a blocking portion surrounding the interface member is formed at an intersection of the interface member and the housing, so as to solve the above problems effectively.

[0030] A card reader provided by the present disclosure will be described in detail below by referring to the accompanying drawings and embodiments.

[0031] The present disclosure provides a connection structure. As shown in FIG. 3, FIG. 3 is a structural schematic view of the connection structure according to an embodiment of the present disclosure. The connection structure **10** of the present disclosure may be configured in a portable storage device, such as a portable solid-state drive, a portable micro-drive, and so on, serving as a male connector to be connected with an interface of an electronic device, such as a mobile phone **20**. The connection structure **10** may alternatively be configured in a charging cable or a data cable to be connected with the interface of the mobile phone **20**. The present disclosure does not limit an application of the connection structure. In the present embodiment, the connection structure **10** includes a housing **11**, an interface member **12**, and a blocking portion **13**.

[0032] At least a portion of the interface member **12** extends from a first side wall **111** of the housing **11** to an outside of the housing **11** to be connected with the interface of the electronic device. The blocking portion **13** is arranged on the first side wall **111** of the housing **11**. The blocking portion **13** extends from the first side wall **111** to the outside of the housing **11** and surrounds an intersection of the interface member **12** and the first side wall **111**. Specifically, the interface member **12** of the connection structure **10** is configured to be connected with the interface of a device, such as the mobile phone, a computer, and the like, so as to achieve data transmission or charging with the interface of the device. The interface member **12** of the connection structure **10** is longer than the interface member **12** in the art, and the blocking portion **13** is arranged surrounding the intersection of the interface member **12** and the first side wall **111**. In the present embodiment, the device is the mobile phone **20**, as shown in FIG. 4, FIG. 4 is a structural

schematic view of the connection structure being connected with the interface of the mobile phone that is uncovered with the phone case, according to an embodiment of the present disclosure. Since the interface member **12** of the connection structure **10** of the present disclosure is longer than the interface member **12** in the art, when the mobile phone **20** is not covered with the phone case **30**, a portion of the interface member **12** that is not surrounded by the blocking portion **13** is inserted into the interface of the mobile phone **20**, and a portion of the interface member **12** disposed outside the interface of the mobile phone **20** is blocked by the blocking portion **13**. In this way, protection for ESD and EMC is ensured, and a pleased outer appearance is provided. As shown in FIG. 5, FIG. 5 is a structural schematic view of the connection structure being connected with the interface of the mobile phone that is covered with the phone case, according to an embodiment of the present disclosure. The portion of the interface member **12** that is not surrounded by the blocking portion **13** is inserted into the interface of the mobile phone **20**, and a region of the interface member **12** disposed at an opening of the phone case **30** is blocked by the blocking portion **13**. In this way, exposure of the interface member **12** at the opening is avoided. Exposure of the interface member **12**, due to the interface member **12** excessively long, may be avoided; and poor signal connection, due to the interface member **12** excessively short, may be avoided.

[0033] According to the present disclosure, the connection structure **10** is provided. The interface member **12** of the connection structure **10** is longer than interface member **12** in the art, and the blocking portion **13** is arranged surrounding the intersection of the interface member **12** and the first side wall **111**. When the connection structure **10** is connected to the mobile phone **20** covered with the phone case **30**, the portion of the interface member **12** that is not surrounded by the blocking portion **13** is inserted into the interface of the mobile phone **20**, and the region of the interface member **12** disposed at the opening of the phone case **30** is covered by the blocking portion **13**. In this way, exposure of the interface member **12** at the opening is prevented. Exposure of the interface member **12**, due to the interface member **12** excessively long, may be avoided; and poor signal connection, due to the interface member **12** excessively short, may be avoided.

[0034] Further, in the present embodiment, the blocking portion **13** surrounds the intersection of the interface member **12** and the first side wall **111**, such that the blocking portion **13** provides a more effective blocking effect, the portion of the interface member **12** disposed at the opening of the phone case **30** is completely blocked, preventing the exposure of the interface member **12**. In some embodiments, the blocking portion **13** may be arranged to attach to at least a portion of the intersection of the interface member **12** and the first side wall **111**. That is, the blocking portion **13** does not completely surround the interface member **12**, and in this case, the blocking portion **13** still prevents the exposure of the interface member **12**.

[0035] In the present embodiment, a height of the blocking portion **13** that extends towards the outside of the housing **11** is not less than 1.6 millimeters. In this way, the height of the blocking portion **13** greater than or equal to a thickness of the phone case **30**, such that the portion of the interface member **12** not surrounded by the blocking portion **13** can be completely inserted into the interface of the mobile phone **20**, ensuring connection reliability. The height of the blocking portion **13** may be any proper value, such as 1.6 millimeters, 1.7 millimeters, 2 millimeters, 2.5 millimeters, and the like, which is not limited herein. The height of the blocking portion **13** may be adaptively adjusted based on the thickness of the phone case **30**. For example, in some embodiments, when the thickness of the phone case **30** is 2 millimeters, the height of the blocking portion **13** is not less than 2 millimeters; when the thickness of the phone case **30** is 2.5 millimeters, the height of the blocking portion **13** is not less than 2.5 millimeters. In some embodiments, the height of the blocking portion **13** extending towards the outside of the housing **11** is 1.6 mm.

[0036] Further, the interface member **12** is one of: a Type-C interface and a lightning interface. Specifically, the Type-C interface and the lightning interface are common interfaces of the mobile phone **20** and can meet requirements of fast charging and high-speed data transmission. In some

embodiments, the interface member **12** may be a mini universal serial bus (mini USB) interface, a micro USB interface, or an interface in other types. A type of the interface is not limited herein. [0037] In the present embodiment, the interface member **12** is the Type-C interface, and the interface member **12** includes a snapping segment **121** that is not surrounded by the blocking portion **13**. An extending height of the snapping segment **121** is not less than 6.5 millimeters. Specifically, the snapping segment **121** is a portion of the interface member **12** that is inserted into the interface of the mobile phone **20**. The height of the snapping segment **121** is configured to be not less than 6.5 millimeters, so as to enable the interface member **12** to have a sufficient length to be inserted into the interface of the mobile phone **20** and to ensure connection reliability. The height of the snapping segment **121** of the interface member **12** is determined according to industrial design specifications for Type-C interfaces. The height of the snapping segment **121** shall be greater than or equal to a depth of the interface of the mobile phone **20** and is less than a sum of the depth of the interface of the mobile phone **20** and the thickness of the phone case **30**. For example, the height of the snapping segment **121** may be any proper value such as 6.5 mm, 6.7 mm, 7 mm, and the like.

[0038] Further, in the present embodiment, a width of the blocking portion **13** is less than **12** millimeters. Specifically, a width direction is a direction A-a as shown in FIG. 3. The width of the blocking portion **13** is configured to be less than **12** mm, so as to enable the blocking portion **13** to be inserted into the opening of the phone case **30** to ensure that the snapping segment **121** of the blocking portion **13** to be inserted into the interface of the mobile phone **20**. In addition, the width of the blocking portion **13** is greater than the width of the interface member **12** to enable the blocking portion **13** to provide the blocking effect. The width of the blocking portion **13** may be 11.9 mm, 11.5 mm, 11 mm, and so on, which is not limited herein. In the present embodiment, the width of the blocking portion **13** is 11.5 mm.

[0039] Further, in the present embodiment, a size of the blocking portion **13** is progressively reduced in a direction extending towards the outside of the housing **11**. Specifically, as shown in FIG. 3, the size of the blocking portion **13** is progressively reduced in an extending direction of the interface member **12**, and an end of the blocking portion **13** that is inserted into the opening of the phone case **30** has a smallest size. In this way, when the interface member **12** is inserted, the blocking portion **13** provides guidance. When the end of the blocking portion **13** that is inserted into the opening of the phone case **30** has the smallest size, the blocking portion **13** may be better aligned with the opening of the phone case **30**, facilitating the blocking portion **13** to be conveniently connected to the interface of the mobile phone **20**.

[0040] Further, in the present embodiment, a second side wall **132** of the blocking portion **13** is connected to a surface **131** of the blocking portion **13** away from the housing **11** through a curved section. In this way, an edge of the blocking portion **13** is curved, and a corner is smooth, and the connection structure **10** has an aesthetic outer appearance, the user may have a better touch feeling. In some embodiments, the second side wall **132** of the blocking portion **13** and the surface **131** of the blocking portion **13** away from the housing **11** may be connected to each other at a right angle. The connection therebetween is not limited herein. In some embodiments, a cross section of the blocking portion **13**, taken along an outward direction along the first side wall **111** of the housing **11**, may be trapezoidal. For example, the cross section of the blocking portion **13**, taken along a plane that is perpendicular to the surface **131** of the blocking portion **13** away from the first side wall **111**, is trapezoidal. That is, connection between the second side wall **132** of the blocking portion **13** and the surface **131** of the blocking portion **13** away from the housing **11** may be a plane. In some embodiments, the cross section of the blocking portion **13**, taken along the outward direction along the first side wall **111** of the housing **11**, may be trapezoidal oval, rectangular, rounded rectangular, and so on, which is not limited herein, as long as the blocking portion **13** surrounds the intersection of the interface member **12** and the first side wall **111**.

[0041] In the present embodiment, the blocking portion **13** and the housing **11** are configured as an

integral and one-piece structure. The blocking portion **13** is formed by the first side wall **111** of the housing **11** protruding outwardly. Specifically, the blocking portion **13** and the housing **11** may be formed by integrally injection molding. By configuring the blocking portion **13** and the housing **11** as the integral and one-piece structure, the connection structure **10** is more reliable and structurally stable, and is likely to be damaged. In some embodiments, the blocking portion **13** and the housing **11** may be manufactured separately, and then the blocking portion **13** and the housing **11** are snapped to each other, and in this way, the blocking portion **13** and the housing **11** are structures that are separable from each other.

[0042] Further, in some embodiments, the blocking portion **13** may be a flexible structure, such as a rubber blocking portion **13**, a flexible plastic blocking portion **13**, and the like. In this way, the blocking portion **13** is prevented from damaging the interface of the mobile phone **20**, such that protection is provided.

[0043] According to the present disclosure, the connection structure **10** is provided. The interface member **12** of the connection structure **10** is longer than interface member **12** in the art, and the blocking portion **13** is arranged surrounding the intersection of the interface member **12** and the first side wall **111**. When the connection structure **10** is connected to the mobile phone **20** covered with the phone case **30**, the portion of the interface member **12** that is not surrounded by the blocking portion **13** is inserted into the interface of the mobile phone **20**, and the region of the interface member **12** disposed at the opening of the phone case **30** is covered by the blocking portion **13**. In this way, exposure of the interface member **12** at the opening is prevented. Exposure of the interface member **12**, due to the interface member **12** excessively long, may be avoided; and poor signal connection, due to the interface member **12** excessively short, may be avoided.

[0044] Correspondingly, the present disclosure further provides the portable storage device, the portable storage device includes a body portion and the connection structure as described in the above embodiments. The body portion is disposed inside the housing of the connection structure.

[0045] The above shows only embodiments of the present disclosure, and does to limit the scope of the present disclosure. Any equivalent structure or equivalent principle transformation, which are performed based on the contents of the specification and the accompanying drawings of the present disclosure and are applied directly or indirectly in other related fields, shall be equivalently included in the scope of the present disclosure.

Claims

1. A connection structure, comprising: a housing; an interface member, wherein at least a portion of the interface member extends from a first side wall of the housing to an outside of the housing to be connected with an interface of an electronic device; and a blocking portion, wherein the blocking portion is arranged on the first side wall of the housing, the blocking portion extends from the first side wall towards the outside of the housing and surrounds an intersection of the interface member and the first side wall.
2. The connection structure according to claim 1, wherein a height of the blocking portion extending towards the outside of the housing is not less than 1.6 millimeters.
3. The connection structure according to claim 2, wherein the height of the blocking portion extending towards the outside of the housing is 1.6 millimeters.
4. The connection structure according to claim 1, wherein the interface member is one of: a Type-C interface, a lightning interface, a Micro USB interface, and a MiniUSB interface.
5. The connection structure according to claim 4, wherein the interface member is the Type-C interface, the interface member comprises a snapping segment that is not surrounded by the blocking portion, the snapping segment has an extension height of not less than 6.5 millimeters.
6. The connection structure according to claim 5, wherein a height of the snapping segment is greater than or equal to a depth of the interface of the electronic device and is less than a sum of the

depth of the interface of the electronic device and a thickness of a protective case that covers the electronic device.

7. The connection structure according to claim 1, wherein a width of the blocking portion is less than 12 millimeters.

8. The connection structure according to claim 1, wherein a size of the blocking portion is progressively reduced along a direction of extending towards the outside of the housing.

9. The connection structure according to claim 8, wherein a cross section of the blocking portion, taken along a plane that is perpendicular to a surface of the blocking portion away from the first side wall, is trapezoidal.

10. The connection structure according to claim 8, wherein connection between a second side wall of the blocking portion and a surface of the blocking portion away from the housing is a curved surface.

11. The connection structure according to claim 1, wherein the blocking portion and the housing are configured as an integral and one-piece structure; the blocking portion is formed by a portion of the first side wall of the housing protruding outwardly.

12. A portable storage device, comprising: a body portion; and a connection structure, wherein the connection structure, comprises: a housing; an interface member, wherein at least a portion of the interface member extends from a first side wall of the housing to an outside of the housing to be connected with an interface of an electronic device; and a blocking portion, wherein the blocking portion is arranged on the first side wall of the housing, the blocking portion extends from the first side wall towards the outside of the housing and surrounds an intersection of the interface member and the first side wall; and the body portion is arranged inside the housing of the connection structure.

13. The portable storage device according to claim 12, wherein a height of the blocking portion extending towards the outside of the housing is not less than 1.6 millimeters.

14. The portable storage device according to claim 13, wherein the height of the blocking portion extending towards the outside of the housing is 1.6 millimeters.

15. The portable storage device according to claim 12, wherein the interface member is one of: a Type-C interface, a lightning interface, a Micro USB interface, and a MiniUSB interface.

16. The portable storage device according to claim 15, wherein the interface member is the Type-C interface, the interface member comprises a snapping segment that is not surrounded by the blocking portion, the snapping segment has an extension height of not less than 6.5 millimeters.

17. The portable storage device according to claim 12, wherein a width of the blocking portion is less than 12 millimeters.

18. The portable storage device according to claim 12, wherein a size of the blocking portion is progressively reduced along a direction of extending towards the outside of the housing.

19. The portable storage device according to claim 18, wherein connection between a second side wall of the blocking portion and a surface of the blocking portion away from the housing is a curved surface.

20. The portable storage device according to claim 12, wherein the blocking portion and the housing are configured as an integral and one-piece structure; the blocking portion is formed by a portion of the first side wall of the housing protruding outwardly.
